

Worksheet

Class XII

Subject: IP (065)

Topic: Data Handling using Pandas – Series

Worksheet#: IP065U1WS001

1. State **True** or **False**: Series is a one-dimensional labeled array capable of holding any data type.
2. State **True** or **False**: Element-wise arithmetic operations (like addition and subtraction) can be performed between two Series of the same length.
3. State **True** or **False**: Series based head() and tail () functions can be used on one single series.
4. State **True** or **False**: Suppose **L1** is list and **S1** is series then **L1*2** (list) and **S1*2** (Series) will give same output?
5. Series is size _____ and value _____.
 - (a) mutable, mutable
 - (b) immutable, immutable
 - (c) mutable,immutable
 - (d) immutable, mutable
6. Series operations align data based on _____ rather than position.
7. To access the first 5 elements of a Series **s**, we use **s._____()**.
8. Which statement creates a Series from a dictionary {'x':100,'y':200}?
 - (a) **pd.Series(['x', 'y'], [100, 200])**
 - (b) **pd.Series({'x': 100, 'y': 200})**
 - (c) **pd.Series(100, 200, index=['x', 'y'])**
 - (d) **pd.Series.from_dict({'x': 100, 'y': 200})**
9. Which of the following is TRUE about a Pandas Series?
 - (a) It is always two-dimensional
 - (b) It has labeled indices
 - (c) It cannot store string values
 - (d) It requires all values to be unique
10. What will **s1 + s2** do when **s1** and **s2** are Series?
 - (a) Concatenate the Series
 - (b) Add elements by position
 - (c) Add elements by matching index labels
 - (d) Subtract elements from each other
11. To display third Element of a Series object **S**, you will write:
 - (a) **S[:3]**
 - (b) **S[2]**
 - (c) **S[3]**
 - (d) **S[:2]**

Assertion(A) and Reason(R) based questions. Mark the correct choice as:

- (a) Both A and R are true and R is the correct explanation for A
- (b) Both A and R are true and R is not the correct explanation for A
- (c) A is True but R is False
- (d) A is False but R is True

12. **Assertion (A):** In a Series, each element must be of the same data type.
Reason (R): A Series in Pandas is similar to a list in Python, which also enforces uniform data types.
13. **Assertion (A):** If an index label is missing in one Series, Pandas will raise an error during operations.
Reason (R): Pandas returns NaN when a label is missing in any Series during operations.
14. **Assertion (A):** Boolean indexing **s[s > 5]** can return a Series with fewer elements.
Reason (R): It filters elements based on a condition.
15. Find the error(s):
 - (a) **import pandas as pd**
s = pd.Series(1, 2, 3)
print(s)
 - (b) **import pandas as pd**
s = pd.Series([1, 2, 3], index=[0, 1, 2])
s[3] = 4
 - (c) **import pandas as pd**
s = pdd.series([10, 20, 30], index=['a', 'b', 'c', 'd'])
print(s[s > '20'])
16. Write the output:
 - (a) **import pandas as pd**
s = pd.Series([5, 10, 15], index=['a', 'b', 'c'])
print(s['b'])

- (b) `import pandas as pd`
`s = pd.Series([10, 20, 30, 40], index=['a', 'b', 'c', 'd'])`
`print(s['b'] + s['d'])`
- (c) `import pandas as pd`
`nums=pd.Series([9,8,7,6,5],index=range(0,10,2))`
`print(nums[1:3])`
- (d) `import pandas as pd`
`s = pd.Series(6,index=range(0,5))`
`print(s)`
- (e) `import pandas as pd`
`import numpy as np`
`data = 'one':'a','two':'b','three':'c'`
`s=pd.Series(data)`
`print(s)`
`print(s.size)`
- (f) `s2=pd.Series(["i","am", "a","student"])`
`print(s2)`
- (g) Consider a given series, **Series1**:

201	700
202	700
203	700
204	700

- Write a program in Python Pandas to create the series and display it.
- (h) Given the following **Series1**:

A	100
B	200
C	300
D	400
E	500

Write the command to create above Series and then double the value in series and store in another series named **Series2**.

- (i) Consider the following Series object, **s**

IP	95
Physics	89
Chemistry	92
Math	95

- Write the Python syntax which will display only IP.
 - Write the Python syntax to increase marks of all subjects by 10.
- (j) Consider a given series **SQTR**:

QTR1	50000
QTR2	65890
QTR3	56780
QTR4	89000
QTR5	77900

Write a program in Python Pandas to create and display the series.

- (k) What will be the output produced by the following programming statements 1 & 2?

```
import pandas as pd
S1=pd.Series(data=[31,41,51])
print(S1>40)      #Statement1
```

print(S1[S1>40]) #Statement2

- (l) Given two series **S1** and **S2**:

S1		S2	
A	39	A	10
B	41	B	10
C	42	D	10
D	44	F	10

Find the output for following python pandas statements?

- S1[:2]*100**
- S1*S2**
- S2[::-1]*10**

- (m) Given the following Series **S1** and **S2**:

S1		S2	
A	10	A	5
B	20	B	4
C	30	C	6
D	40	D	8

Write the command to find the multiplication of series **S1** and **S2**.

- (n) Consider a given Series, **Subject**:

ENGLISH	75
HINDI	78
MATHS	82
SCIENCE	86

Write a program in Python Pandas to create this series.

- (o) Consider the following Series object, “company” and its profit in Crores:

TCS	350
Reliance	200
L&T	800
Wipro	150

- Write the command which will display the name of the company having profit>250.
- Write the command to name the series as Profit.

- (p) Consider two objects **a** and **b**. **a** is a list whereas **b** is a Series. Both have values **10,20,25,50**. What will be the output of the following two statements considering that the above objects have been created already:

- print(a*2)**
- print(b*2)**

Justify your answer.

- (q) If series **s1** is having following data:

1	6
3	1
5	3
7	5
9	4
11	8
13	7
15	4
17	6
19	7

dtype: int64

What would be the result of the command **print(s1[3:6])**?

- (r) What will be the output of the following code?


```
import pandas as pd
import numpy as np
s = pd.Series(np.arange(10,50,10))
print(s)
print(s.ndim)
print(s.shape)
print(len(s))
```

(s) Consider the following Series s –

```
0    4.0
1    5.0
2    7.0
3    NaN
4    1.0
5   10.0
dtype: float64
```

- Write a Python code to add 1 to all the elements.
- Write a code to replace all NaN with 0.

17. Complete the given Python code to get the required output as: Rajasthan

```
import _____ as pd
di = {'Corbett': 'Uttarakhand', 'Sariska': 'Rajasthan', 'Kanha': 'Madhya Pradesh', 'Gir': 'Gujarat'}
NP = _____. Series( _____ )
print(NP[ _____ ])
```

18. Complete the given Python code to get the required output as "California":

```
import _____ as pd
data = {'Yosemite': 'California', 'Yellowstone': 'Wyoming', 'Glacier': 'Montana', 'Rocky Mountain': 'Colorado'}
national_parks = pd.Series(_____ )
print(national_parks _____)
```

19. Predict the output of the code:

```
import pandas as pd
list1=[-10,-20,-30]
ser = pd.Series(list1*2)
print(ser)
```

20. Monthly Car Sales of 12 months for a Company are given in the form of List:

L = [878, 1293, 7564, 9234, 1012, 3493, 9003, 10021, 3005, 999, 848, 3202]

Answer the following questions:

- Write python code to create a Series CarSales using the above list and give names of months as index for the series.
- Write Python code to print the month and number of cars if the sale for the month is > 5000.